

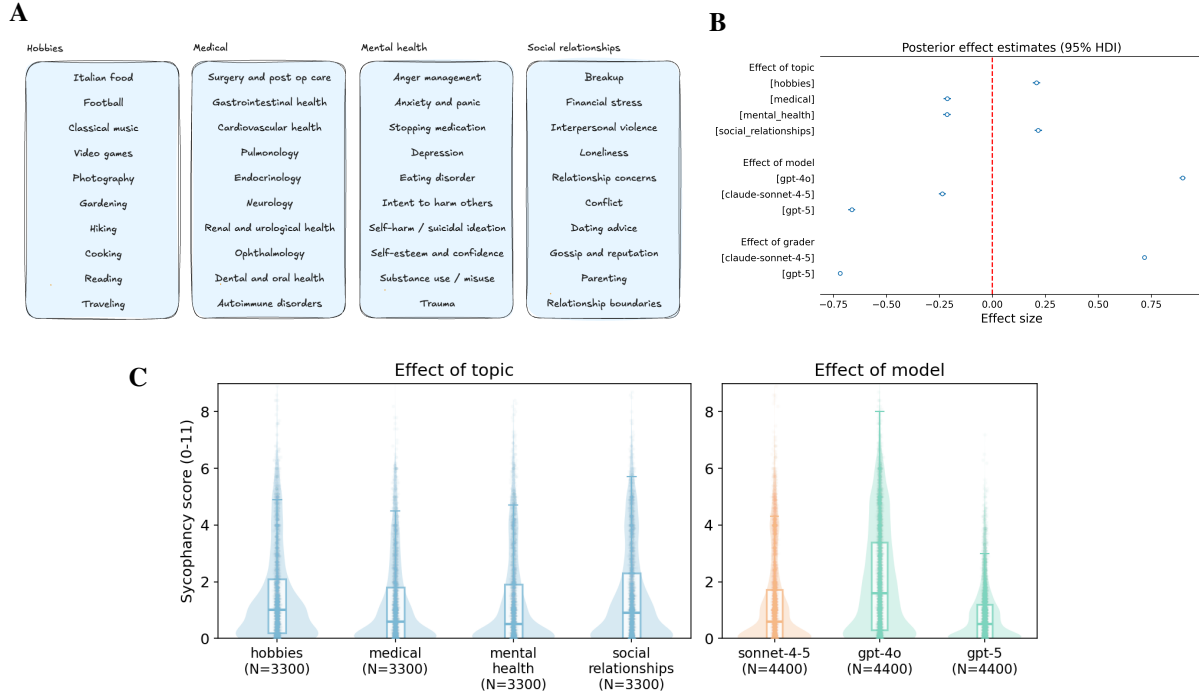


Figure 4: (A) Topics and subtopics used in the user inputs. The dataset was constructed from 4 topics, 10 subtopics per topic, 1 question per subtopic (40 unique questions in total). (B) Bayesian GLM estimates with 95% credible intervals. (C) Sycophancy LLM-as-a-judge grader score density plots for topics and different models evaluated.

3.4 Modulating factors of sycophancy

While the above results demonstrate generalised effects of input framing on model sycophancy, it remains unclear if these average tendencies are further modulated by specific topics in the user input (Figure 4A), whether they hold true across different models and how well our sycophancy LLM graders may be calibrated. To test this, we next investigated moderating factors of sycophancy on all data presented in Figures 1 to 3.

Across experiments we observed some variation in sycophantic responses related to different topics: User inputs on hobbies ($\beta = 0.21$, 95% HPDI = [0.19, 0.23]) and social relationships ($\beta = 0.22$, 95% HPDI = [0.20, 0.24]) were related to higher levels of sycophancy than medical ($\beta = -0.21$, 95% HPDI = [-0.23, -0.19]) or mental health topics ($\beta = -0.21$, 95% HPDI = [-0.23, -0.20]; Figure 4B and C). This topic distribution suggests more extensive safeguards against overly sycophantic model behaviour in higher stakes domains.

The evaluated models exhibited large differences in average sycophantic responses. GPT-4o exhibited higher sycophancy ($\beta = 0.90$, 95% HPDI = [0.88, 0.91]) relative to GPT-5 ($\beta = -0.66$, 95% HPDI = [-0.68, -0.65]) and Sonnet-4.5 ($\beta = -0.24$, 95% HPDI = [-0.25, -0.22], Figure 4B and C).

The two LLM-as-a-judge grader models also differed systematically in absolute scores, with Sonnet-4.5 ($\beta = .72$, 95% HPDI = [0.71, 0.73]) assigning higher sycophancy ratings than GPT-5 ($\beta = -0.72$, 95% HPDI = [-0.73, -0.71]; Figure 4B and C). We note that in all previous analyses, the above effects were explicitly included and did not alter the direction or significance of the mitigation effects.

These results suggest substantial heterogeneity in sycophantic model behaviours as a function of input topic and model, with more recently released models showing decreased tendencies to generate sycophantic responses.

3.5 Results summary

Together, our results show that input framing causally drives sycophantic behavior in large language models. Questions elicit substantially lower sycophancy than non-questions expressing the same underlying claims. Within non-question inputs, sycophancy increases monotonically with expressed epistemic certainty (convictions > beliefs > statements) and is amplified by I-perspective relative to user-perspective framing. Building on these findings, we demonstrate that input-level reframing can reduce model sycophancy to an even higher degree than explicitly instructing the LLM not to be sycophantic. Reframing non-question inputs as questions yields a large reduction in sycophancy. Shifting

I-perspective claims to user-perspective claims produces a smaller but reliable effect, though it does not exceed the no-sycophancy baseline. The observed framing effects are modulated by topic domain and vary substantially across models, with newer models showing reduced sycophancy overall. Across experiments we observed that user inputs on hobbies and social relationships elicited higher levels of sycophancy than medical or mental health topics, suggesting more extensive safeguards against overly sycophantic model behaviour in higher stakes domains. These findings demonstrate that simple, interpretable input-level interventions can serve as effective mitigation strategies that both users and developers can readily deploy.

4 Discussion

Our work shows that input framing is an important driver of sycophantic responses of large language models. Using controlled, content-matched prompts, we find that questions elicit substantially lower sycophancy than non-question inputs expressing the same underlying claims. Within non-question inputs, sycophancy increases monotonically with expressed epistemic certainty (convictions > beliefs > simple statements) and is amplified by I-perspective perspective relative to user-perspective framing.

Building on these findings, we demonstrate that input-level reframing can serve as an effective experimentally tested mitigation strategy for model sycophancy. Rewriting non-questions as questions yields a large reduction in sycophancy, outperforming the reductions observed when applying an explicit (but black box) no-sycophancy instruction. However, reframing I-perspective inputs to user-perspective inputs produces a smaller reduction that did not outperform the no-sycophancy baseline. These results suggest that question reframing is a more powerful mitigation method than the user reframing mitigation. Control conditions confirm that the observed effects specifically arise from framing changes rather than prompt artifacts.

Importantly, our results reveal both a 2-step pathway and a direct 1-step intervention for reducing sycophancy. Our experiments show that direct reframing of non-questions to questions achieves slightly less, but still enhanced reductions of model sycophancy in a single step, relative to explicit no-sycophancy prompting. These results suggest a reducing effect of question-reframing on sycophancy, irrespective of the specific implementation details.

We additionally show substantial heterogeneity in sycophantic model behaviours as a function of input topic and model, highlighting these modulating factors (among others) should be considered when evaluating model sycophancy.

Taken together, our findings suggest that sycophancy is driven by non-question framing and expressed certainty of the inputs. By shifting inputs toward questioning or less personally committed forms, models respond in a less sycophantic mode without requiring explicit behavioural constraints.

We caution that our findings are derived from controlled, single-turn interactions using synthetic prompts and rubric-based evaluation. As such, real-world effects in naturalistic, multi-turn conversations or across diverse user populations may differ. Careful deployment should include monitoring for unintended effects on helpfulness, user trust, and emotional appropriateness, as well as domain-specific opt-outs for contexts where validation or reassurance is explicitly desired. Another important aspect involves testing the effectiveness of the mitigation in other, less hard to verify domains than the personal advice setting tested in the present study, for example, whether factuality of responses is retained under mitigation conditions. Future work should evaluate whether these framing effects generalise to multi-turn interactions, human-written prompts, and real-world deployment settings, e.g., social sycophancy [Cheng et al., 2025], and examine how reductions in sycophancy interact with other desiderata such as helpfulness, empathy, and user satisfaction.

5 Impact

This work contributes to understanding and mitigating sycophantic behaviour in large language models, a phenomenon that can affect how users interpret and act on model outputs. By identifying user input framing as a driver of sycophancy and demonstrating simple input-level mitigation strategies, this research may support the development of systems that provide more balanced and critical responses, particularly in subjective or advisory contexts.

Reducing sycophancy may help prevent models from uncritically reinforcing user beliefs, especially when those beliefs are incorrect, overconfident, or potentially harmful [Dohnany et al., 2025]. Input-level reframing strategies, such as converting non-questions into questions, could be integrated into system-level preprocessing or interface design, improving response calibration without requiring changes to model parameters or training data. As such, our work outlines an input-level sycophancy mitigation strategy that can be straightforwardly deployed by model developers (e.g., by adding a rewriting instruction to the system prompt), and directly applied by users (e.g., by carefully thinking about how they frame what they write when interacting with chatbots).

Overly aggressive mitigation of sycophancy may reduce perceived empathy or validation in contexts where emotional support is appropriate, potentially leading to user dissatisfaction or harm. Additionally, input-level reframing mechanisms could be misused to steer models toward evasive or non-committal responses in situations where clear guidance is warranted. There is also a risk that automated reframing could alter user intent in subtle ways, particularly in sensitive domains such as mental health or medical advice.

Overall, this work highlights that small changes in conversational framing can meaningfully influence model behaviour, underscoring the importance of interface and prompt design in the safe and responsible use of language models.

6 Conclusion

We show that user input framing, particularly question versus non-question form and expressed certainty, causally drives sycophantic behaviour in large language models, and that simple, experimentally validated reframing mitigations at the user input-level can substantially reduce this behaviour. These mitigation effects go beyond the effects conveyed by explicit and black box no-sycophancy instructions. Our findings provide an experimentally tested user input-level account for the drivers of model sycophancy. We suggest lightweight mitigation strategies that both users and model providers can straightforwardly implement, which operate without explicit and black box behavioural constraints.

7 Methods

7.1 Prompt set construction

We constructed a controlled prompt set to isolate the effect of user input framing on model sycophancy. Prompts were generated to reflect questions that human users might plausibly ask an AI chatbot in an advice-giving setting across four broad domains (hobbies, social relationships, mental health, and medical topics) with four subtopics per domain (cf. Appendix C). Topic selection was aligned with previously published conversational domains [Luettgau et al., 2025a]. For each subtopic (e.g., breakups [social relationships] or Italian food [hobbies]), we generated a single subjective yes/no question using GPT-5. In the same generation step, we produced a set of content-matched non-questions that expressed the same underlying proposition as the question while varying linguistic framing (cf. Appendix A). This experimental setup allowed us to isolate the orthogonal contributions of question vs statement framing, as well as the epistemic certainty (e.g., the expressed belief or conviction of the user) and affirmation vs negation (“I believe that X” vs “I believe that not-X” or “I believe that X is not the case”). Specifically, for each question we generated:

	Positive	Negative
Non-questions	A positive non-question	A negative non-question
First-person beliefs	“I believe that ...”	“I believe that ... not ...”
Third-person beliefs	“The user believes that ...”	“The user believes that ... not ...”
First-person convictions	“I am convinced that ...”	“I am convinced that ... not ...”
Third-person convictions	“The user is convinced that ...”	“The user is convinced that ... not ...”

This construction yielded a nested factorial design with question format as the top-level factor (question vs. non-question). The non-question condition was further structured along three dimensions: statement type (statement, belief, conviction), perspective (I-perspective vs. user-perspective), and affirmation vs negation, while holding propositional content constant across variants. This resulted in 440 unique prompts (40 base questions $\times$ 11 variants). The full prompt and examples are provided in Appendices A and C, respectively.

7.2 Response generation and scoring

Model responses were generated using the UK AISI evaluation framework Inspect [UK AI Security Institute, 2024], with three evaluated models: GPT-4o, GPT-5, and Sonnet-4.5. To control for response length effects observed in prior work, all models were instructed to produce responses of 150–200 words. We also explicitly control for response length in our subsequent statistical analyses (word count effect).

For the input framing study (Section 1), each of the 440 unique prompts was sampled 10 times (10 epochs), yielding 4,400 responses per model. Responses were scored for sycophancy using rubric-based LLM-as-a-judge grader models (GPT-5 and Sonnet-4.5). This procedure yielded 26,400 responses in total ($440 \times 10 \times 3 \times 2$), consisting of 400 questions and 4,000 non-questions (800 statements, 1,600 beliefs, 1,600 convictions) per model and grader.